

PERMUTATION AND COMBINATION

Directions for question 1: Type in your answer in the input box provided in the question.

If ${}^nC_4 = {}^nC_9$, the value of ${}^nC_{11}$ is

Solution:

$$\text{If } {}^nC_r = {}^nC_s \Rightarrow r = s \text{ or } n = r + s$$

$$\therefore r = 4 \text{ and } s = 9$$

$$\Rightarrow n = 9 + 4 = 13$$

$${}^nC_{11} = {}^{13}C_{11} = 78.$$

Ans : (78)

Directions for question 2: Select the correct alternative from the given choices.

If ${}^nC_r = 220$ and ${}^nP_r = 1320$, the value of nP_2 is _____ .

- ☐ a) 120
- ☐ b) 116
- ☐ c) 136
- ☐ d) 132

Solution:

$$\frac{{}^nP_r}{{}^nC_r} = r!$$

$$\Rightarrow r! = \frac{1320}{220} = 6$$

$$r! = 3! \Rightarrow r = 3$$

$${}^nP_r = 1320$$

$$\text{i.e. } {}^nP_3 = 1320 = 12 \times 11 \times 10$$

$$\therefore n = 12$$

$${}^nP_2 = {}^{12}P_2 = 12 \times 11 = 132$$

Choice (D)

Directions for question 3: Type in your answer in the input box provided in the question.

$${}^{15}C_6 + {}^{15}C_8 + {}^{15}C_{10} - {}^{15}C_9 - {}^{15}C_7 - {}^{15}C_5 =$$

Solution:

$${}^{15}C_6 = {}^{15}C_9 \quad (\because {}^nC_r = {}^nC_{n-r})$$

$${}^{15}C_8 = {}^{15}C_7 \text{ and } {}^{15}C_{10} = {}^{15}C_5$$

$$\therefore {}^{15}C_6 + {}^{15}C_8 + {}^{15}C_{10} - {}^{15}C_9 - {}^{15}C_7 - {}^{15}C_5 = 0$$

Ans : (0)

Directions for questions 4 to 6: Select the correct alternative from the given choices.

$${}^{18}C_{10} + {}^{18}C_9 = \underline{\hspace{2cm}}.$$

- ☐ a) ${}^{19}C_8$
- ☐ b) ${}^{19}C_{10}$
- ☐ c) ${}^{19}C_{11}$
- ☐ d) None of these

Solution:

We have ${}^nC_r + {}^nC_{r-1} = {}^{(n+1)}C_r$

Using this result

$${}^{18}C_9 + {}^{18}C_{10} = {}^{19}C_{10}$$

Choice (B)

Directions for questions 4 to 6: Select the correct alternative from the given choices.

How many different six letter words that begin with E can be formed using all the letters of the word DEVINE?

- ☐ a) 110
- ☐ b) 120
- ☐ c) 140
- ☐ d) 100

Solution:

Since the word has to begin with E, the first blank can be filled in one way and the remaining five blanks can be filled in $5 \times 4 \times 3 \times 2 \times 1$ ways.
Hence the required number of words is 120

Choice (B)

Directions for questions 4 to 6: Select the correct alternative from the given choices.

How many seven letter words can be formed using any 3 vowels and any 4 consonants?

☐ a) $7!$

☒ b) ${}^5C_3 \times {}^{21}C_4$ (Your answer - ✖)

☐ c) $7! \times {}^5C_3 \times {}^{21}C_4$ (Correct Answer - ✔)

☐ d) ${}^{26}C_7 \times 7!$

Solution:

3 vowels out of 5 can be selected in 5C_3 ways and 4 consonants out of 21 can be selected in ${}^{21}C_4$ ways.
 $\therefore$ The required number of words that can be formed = ${}^5C_3 \times {}^{21}C_4 \times 7!$.

Choice (C)

Directions for question 7: Type in your answer in the input box provided in the question.

There are three places A, B and C in a city such that there are 4 roads connecting A and B and 5 roads connecting B and C. The number of ways in which one can travel from A to C via B is

Solution:

The number of ways one can travel from A to B is 4 and the number of ways one can travel from B to C is 5. $\therefore$ the total number of ways one can travel from A to C via B is $4 \times 5 = 20$.

Ans : (20)

Directions for question 8: Select the correct alternative from the given choices.

How many different words can be formed using all the letters of the word FEBRUARY such that each word ends with a vowel?

- ☐ a) $\frac{7!}{2!}$
- ☐ b) $6! \times 3$

☒ c) $\frac{8!}{2!} \times 3$ (Your answer - ✖)

- ☐ d) $\frac{7!}{2!} \times 3$

Solution:

In the word FEBRUARY there are 3 vowels and 5 consonants of which 2 are R's. As the words formed are to end with a vowel the last letter of the word can be fixed in 3 ways and the remaining 7 places can be filled by the remaining 7 letters in $\frac{7!}{2!}$ ways.

$$\therefore \text{The required number of words} = \frac{7!}{2!} \times 3$$

Choice (D)

Directions for questions 9 and 10: Type in your answer in the input box provided below the question.

Harika wants to invite some of her six friends to a party. In how many ways can she invite them?

Solution:

Out of her six friends she may invite either one or two or three or four or five or six which can be done in ${}^6C_1 + {}^6C_2 + {}^6C_3 + {}^6C_4 + {}^6C_5 + {}^6C_6 = 2^6 - 1 = 63$ ways

Ans : (63)

Directions for questions 9 and 10: Type in your answer in the input box provided below the question.

If Harika wants to invite at least four of her six friends to the party, in how many ways can she do it?

Solution:

If Harika wants to invite at least four of her friends then she can invite either four or five or six friends which can be done in ${}^6C_4 + {}^6C_5 + {}^6C_6$ ways = 22 ways.

Ans : (22)

Directions for questions 11 to 15: Select the correct alternative from the given choices.

If Harika wants to invite at most four of her six friends to the party, in how many ways can she do it?

- ☐ a) 60
- ☐ b) 57
- ☐ c) 36
- ☐ d) 58

Solution:

Harika wants to invite at most 4 persons. It means she wants to invite either zero or one or two or three or four only. This can happen in ${}^6C_0 + {}^6C_1 + {}^6C_2 + {}^6C_3 + {}^6C_4 = 57$ ways

Alternative Solution:

The number of ways of inviting at most 4 friends = total number of ways of inviting – number of ways of inviting more than four of her friends

$$\text{i.e. } 2^6 - ({}^6C_5 + {}^6C_6) \\ = 64 - (7) = 57.$$

Choice (B)

Directions for questions 11 to 15: Select the correct alternative from the given choices.

In how many ways can 10 friends be seated around a circular table such that two particular persons are always to sit together?

- ☒ a) 9! (Your answer - ✖)
- ☐ b) $8! \times 2!$
- ☐ c) $10!$
- ☐ d) $8!$

Solution:

As two particular persons are to sit together let the two persons be treated as one unit. With the remaining 8 we have to arrange 9 units of persons around the circular table. This can be done in $(9 - 1)! \times 2! = 8! \times 2!$ ways

Choice (B)

Directions for questions 11 to 15: Select the correct alternative from the given choices.

Twenty three friends want to go to an educational tour. They are to be accommodated in two vehicles – a 7-seater and a 16-seater. In how many ways can they be accommodated in the two vehicles?

- ☐ a) ${}^{23}C_7$
- ☐ b) ${}^{23}C_7 \times {}^{23}C_{16}$
- ☐ c) ${}^{23}C_{14}$
- ☐ d) ${}^{23}C_{15}$

Solution:

In one vehicle 7 and in the other 16 persons are to be accommodated. This can be done in ${}^{23}C_7$ or ${}^{23}C_{16}$ ways.

Choice (A)

Directions for questions 11 to 15: Select the correct alternative from the given choices.

Along a railway line there are 6 stations between two stations A and B. How many different tickets are to be printed so that a person can travel from any station to any other?

- ☐ a) 28
- ☐ b) 56
- ☐ c) 24
- ☐ d) 36

Solution:

Altogether there are 8 (i.e, $6 + 2$) stations. The number of tickets that are to be printed $= {}^8P_2 = 56$.

Choice (B)

Directions for questions 11 to 15: Select the correct alternative from the given choices.

There are 5 points on a plane. How many triangles can be formed by joining these points?

☒ a) 60 (Your answer - ✖)

☐ b) 10

☐ c) 120

☐ d) Cannot be determined

Solution:

As we do not know whether the given 5 points are collinear or some of them are collinear or none of them are collinear we cannot determine.

Choice (D)